

Project Design Document

01/24/2026
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Project Concept

1 Player Control	You control a		in this	
	<i>Camera</i>		<i>Isometric 3D</i>	game
	where		makes the player	
	<i>Mouse and Keyboard input</i>		<i>pan the camera across the map, zoom in/out to inspect districts, and select grids for construction.</i>	
2 Basic Gameplay	During the game,		from	
	<i>various buildings (Houses, Factories, Villas, Parks)</i>	appear	<i>player selection on empty construction plots</i>	
	and the goal of the game is to			
	<i>grow the city's population to the target level while maintaining a high Eco-Score, balancing expansion with environmental health.</i>			
3 Sound & Effects	There will be sound effects		and particle effects	
	<i>for building construction, UI button clicks, cash collection chimes, and victory/defeat alerts, car engine sounds</i>		<i>showing smoke rising from factories and glowing highlights when hovering over buildable plots.</i>	
	[optional] There will also be			
	<i>autonomous vehicles that follow traffic paths and park themselves, bringing life to the city streets.</i>			
4 Gameplay Mechanics	As the game progresses,		making it	
	<i>buildings automatically generate income and population over time,</i>		<i>crucial to manage resources (Money/Eco-Score) to afford further expansion without ruining the environment.</i>	
	[optional] There will also be			
	<i>a smart traffic system where vehicles autonomously navigate the road network and find parking spots using pathfinding logic.</i>			

5

User Interface

The

*Money,
Population, and
Eco-Score*

will

*increase or
decrease*

whenever

*new buildings are constructed or the city
cycle generates tax income.*

At the start of the game, the title

City Builder

will appear

and the game will end when

*the player reaches the target population
and Eco-Score (Victory) while construction
on all over the map, or runs out of space
(Game Over)*

Project Timeline

Milestone**Description****Due**

#1

*- Free asset research**01/24*

#2

*- UI Design for the title screen and the map in the main scene**01/24*

#3

*- Functional gameplay implementation**01/25*

#4

*- Functional gameplay implementation**01/26*

#5

*- Extra effects (car animations, end game panels)**01/27*

Programming Theory Principles Implementation (OOP Pillars)

- **Abstraction:** "Will be used in the ConstructionPlot system, where the complex logic of building states is hidden behind simple click interactions."
- **Inheritance:** "Will be used for Building types (e.g., Factory, House inherit from a base Building class)."
- **Polymorphism:** "Will be used in the Building calculations (different construction costs, income rates for different building types)."
- **Encapsulation:** "Will be used in the DataManager script, keeping isOccupied private and only allowing changes via specific public methods)."

Project Sketch

Gold: 1000

Population: 30/100

EcoScore: 30/80

